

(56)

References Cited

U.S. PATENT DOCUMENTS

7,503,336 B2 * 3/2009 Labarbera A45B 25/18
160/24
7,779,849 B2 * 8/2010 Labarbera A45B 25/18
135/33.7
7,814,920 B2 * 10/2010 Lenahan A45B 11/00
135/21
8,047,217 B1 11/2011 Schermerhorn, Jr.
8,640,820 B2 2/2014 Caeti et al.
9,057,206 B2 * 6/2015 Lin E04H 15/28
9,538,819 B1 * 1/2017 Hayes A45B 19/06
9,629,428 B1 * 4/2017 Nei A45B 25/18
9,775,416 B2 10/2017 Harris
9,936,811 B2 * 4/2018 Rowe, Jr. E04H 15/02
10,182,627 B2 * 1/2019 Nei A45B 25/02
10,697,196 B2 * 6/2020 Nicholson, Jr. E04H 15/58
10,791,843 B2 * 10/2020 Rowe, Jr. A47C 7/66
10,925,363 B1 * 2/2021 Schermerhorn A45B 23/00

11,612,223 B1 * 3/2023 Schermerhorn A45B 25/02
135/16

2004/0107979 A1 6/2004 Lai
2005/0011134 A1 * 1/2005 Boltan E04H 15/003
52/3
2017/0238661 A1 8/2017 Haythornthwaite
2018/0035770 A1 * 2/2018 LaBella A45B 19/04
2020/0329835 A1 * 10/2020 Ferrara A45B 25/20

FOREIGN PATENT DOCUMENTS

EP 0531240 A1 9/1992
FR 1110111 A * 2/1956
FR 2713265 A1 * 6/1995 A45B 23/00
KR 200404998 Y1 * 1/2006
KR 200448498 Y1 * 4/2010
KR 20140003970 A * 1/2014
KR 20230009565 A * 1/2023
WO WO-2007027979 A2 * 3/2007 A45B 15/00
WO WO-2015019031 A1 * 2/2015 E04F 10/02

* cited by examiner

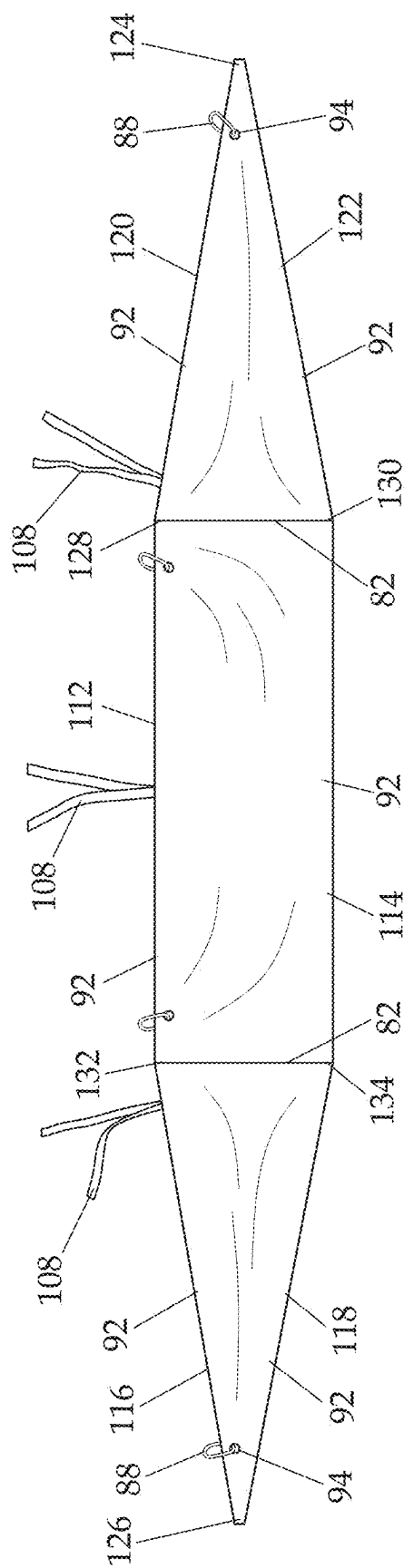


FIG. 1

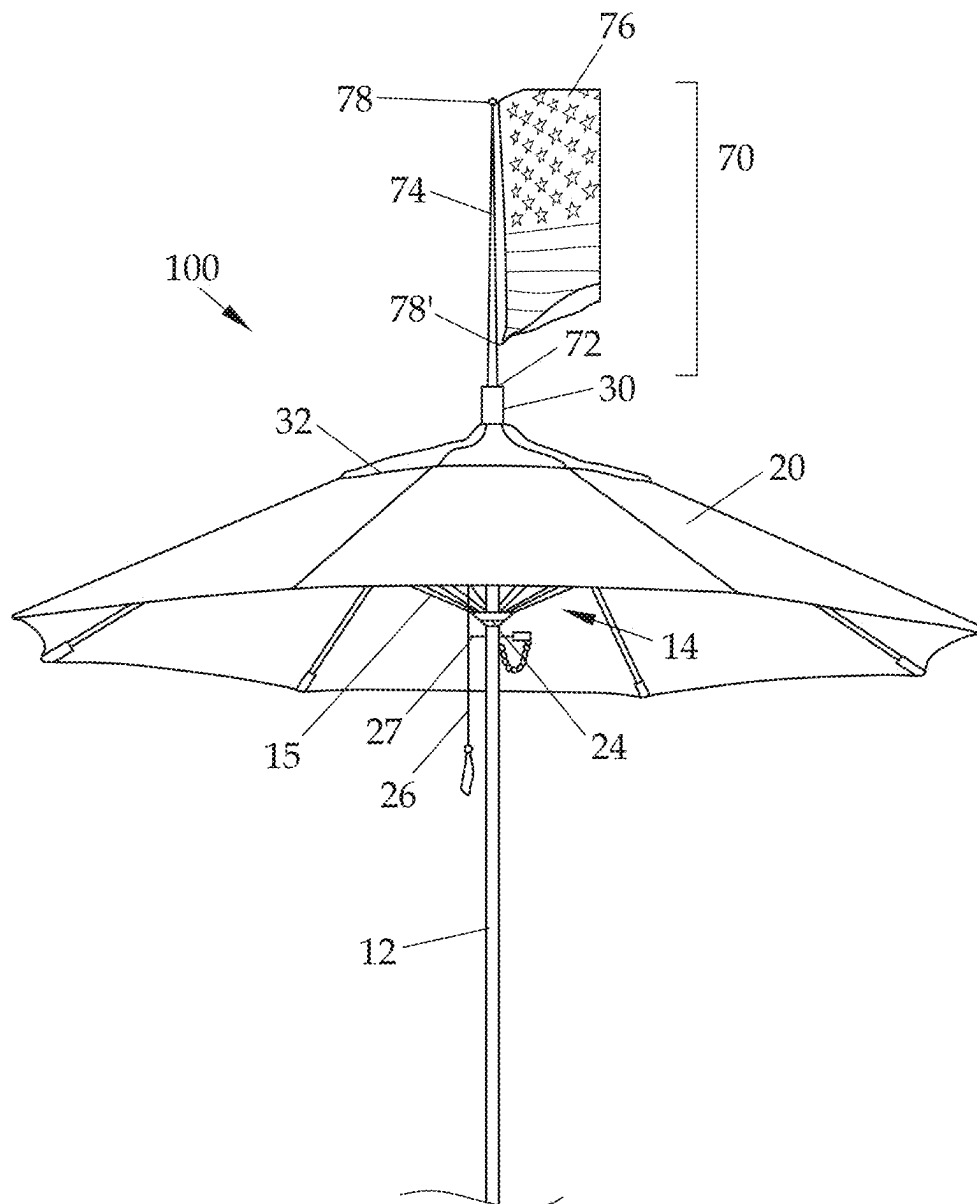


FIG. 1a

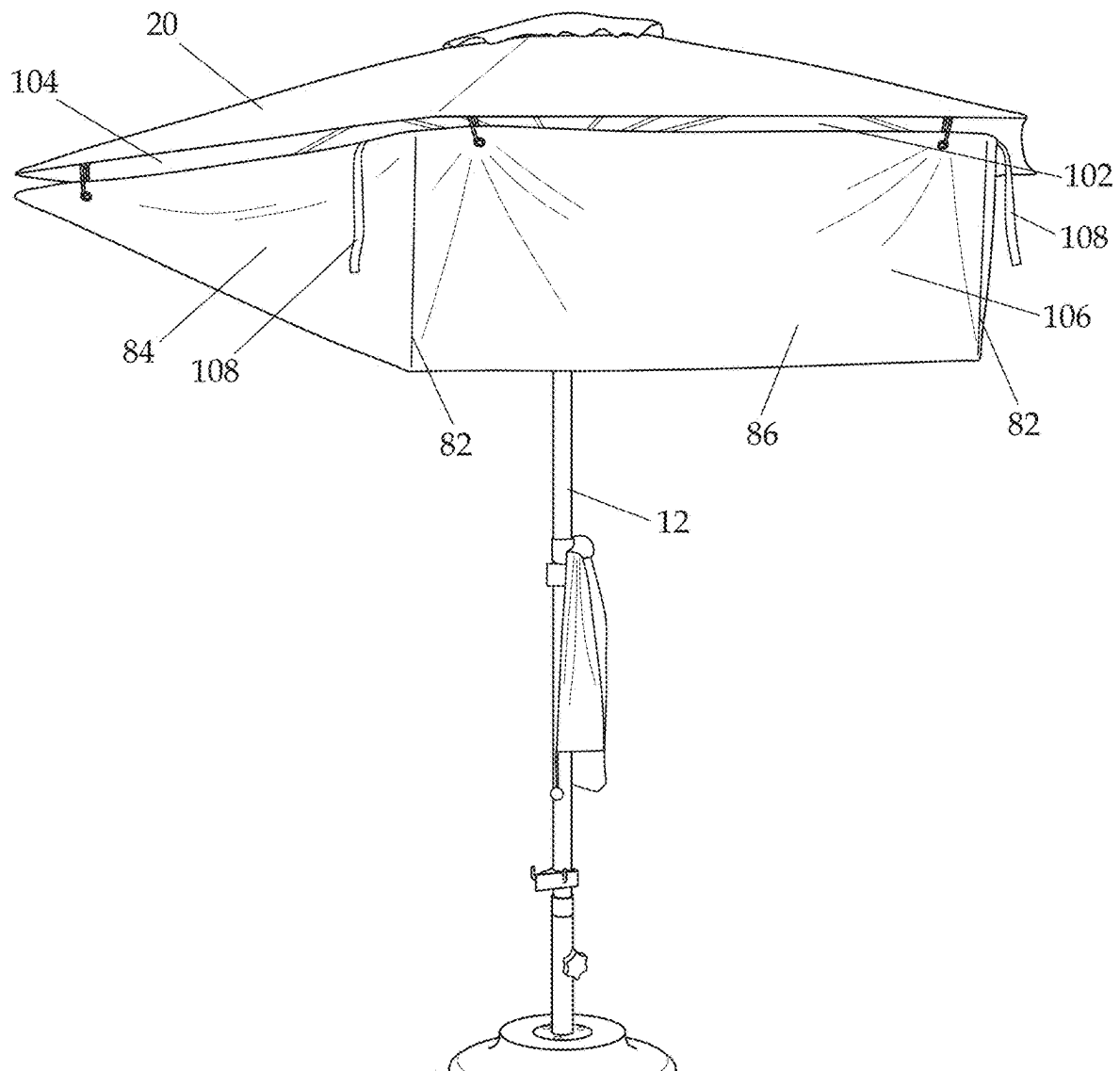


FIG. 1b

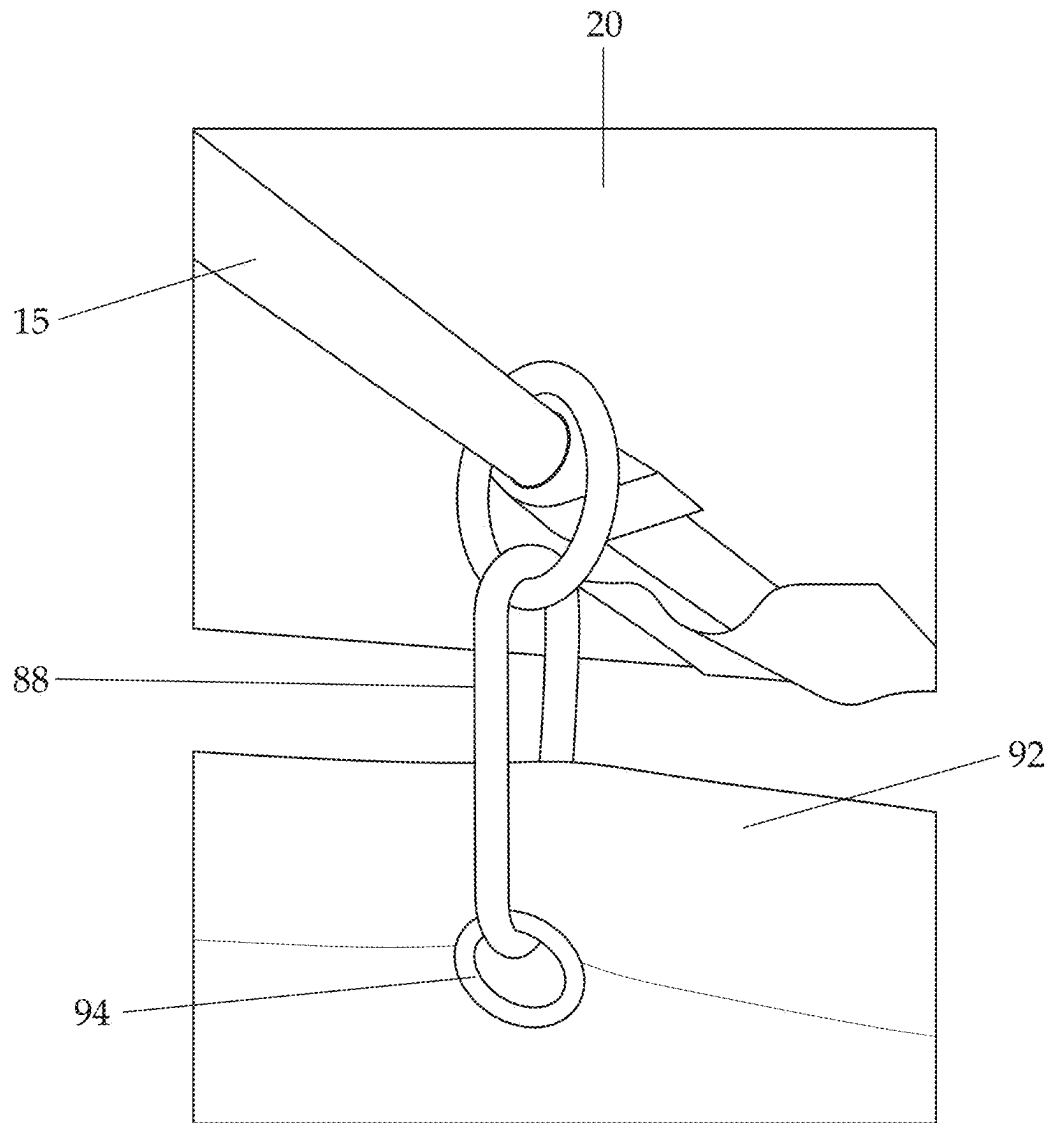


FIG. 1c

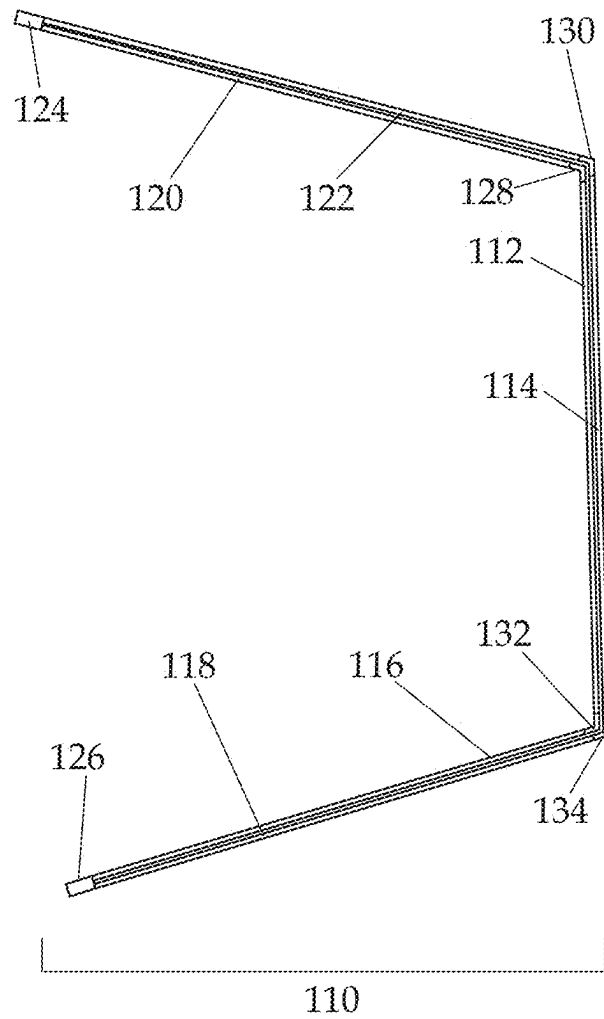


FIG. 1d

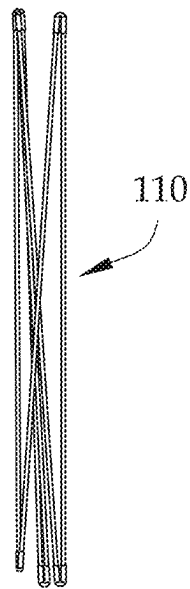


FIG. 1e

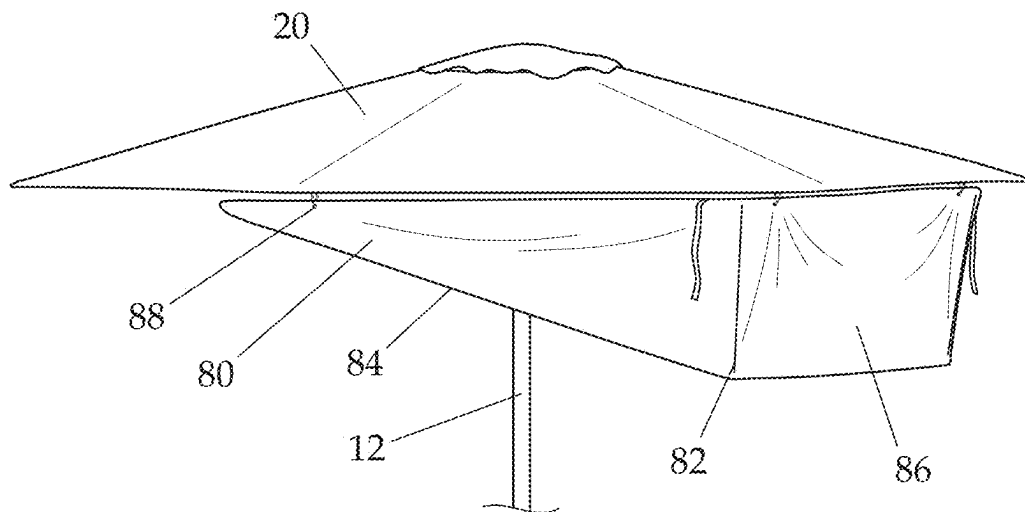


FIG. 1f

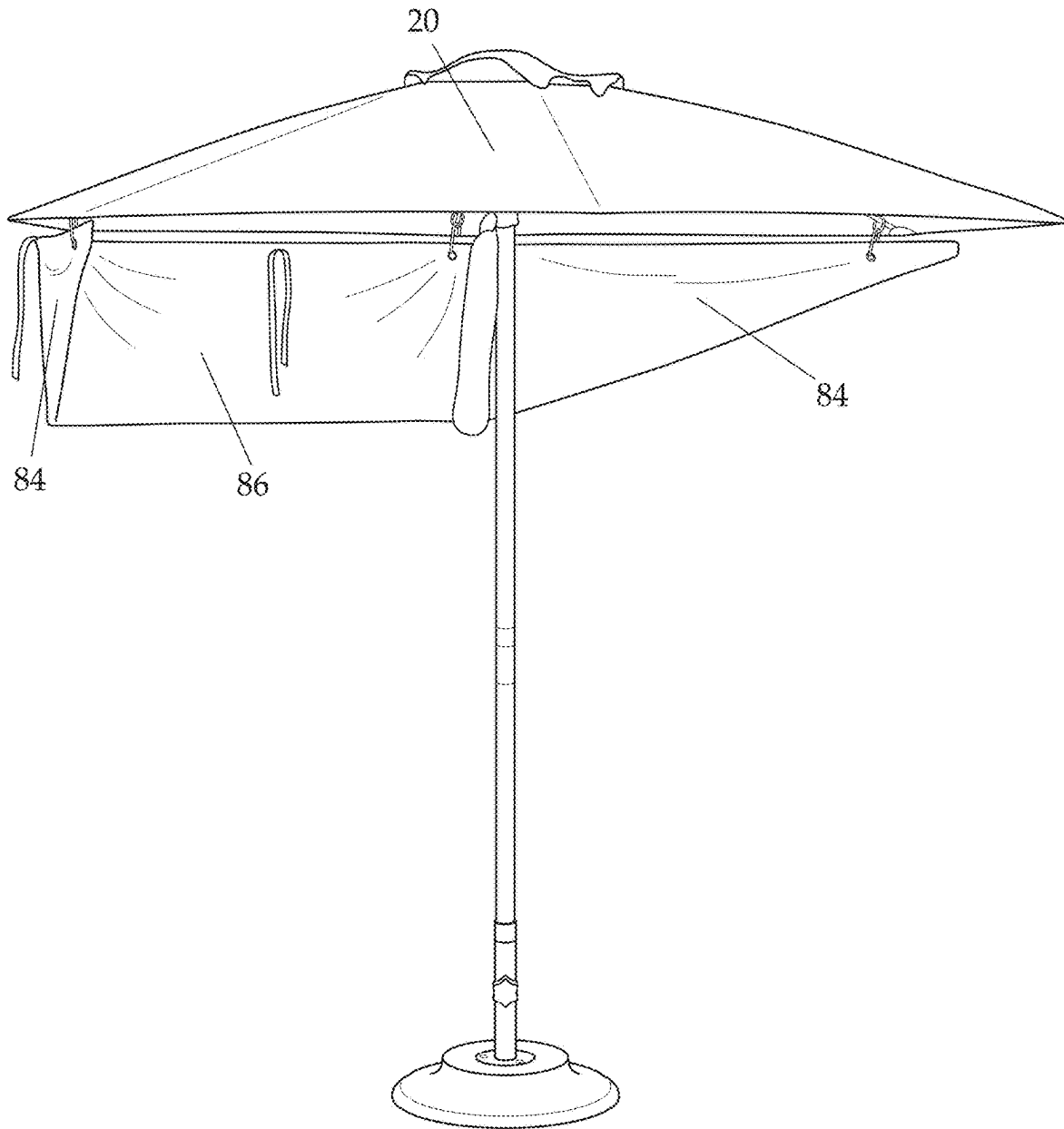


FIG. 1g

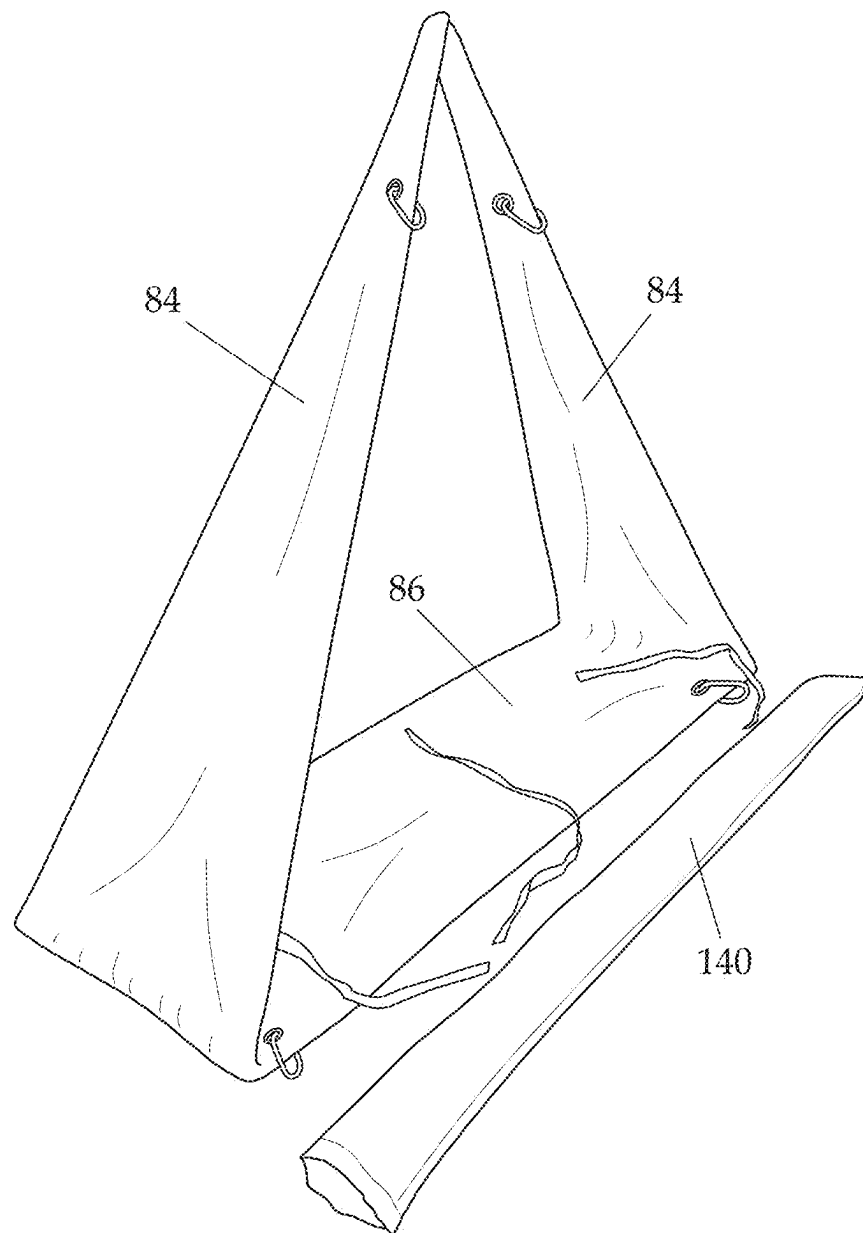


FIG. 1h

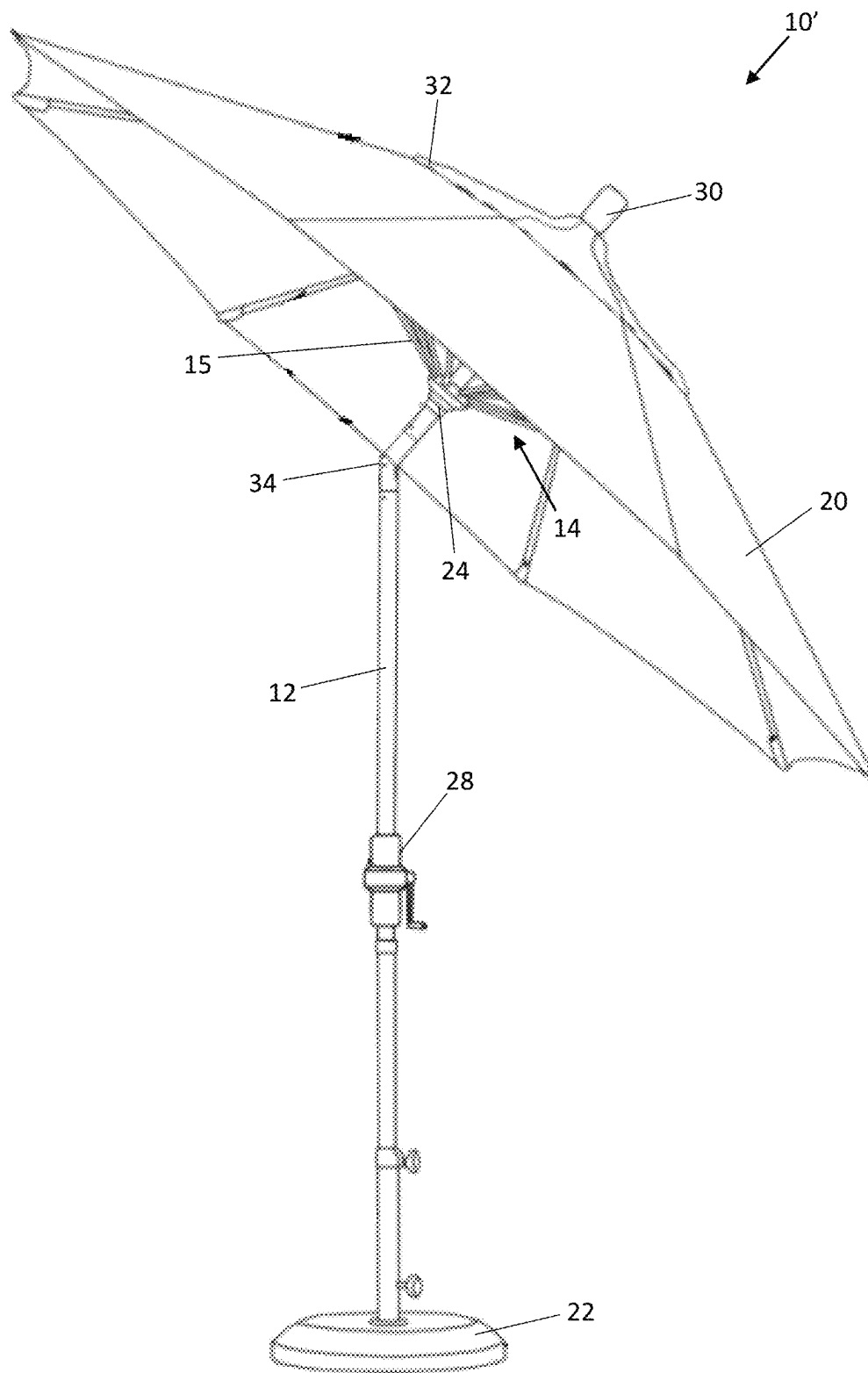


FIG. 2

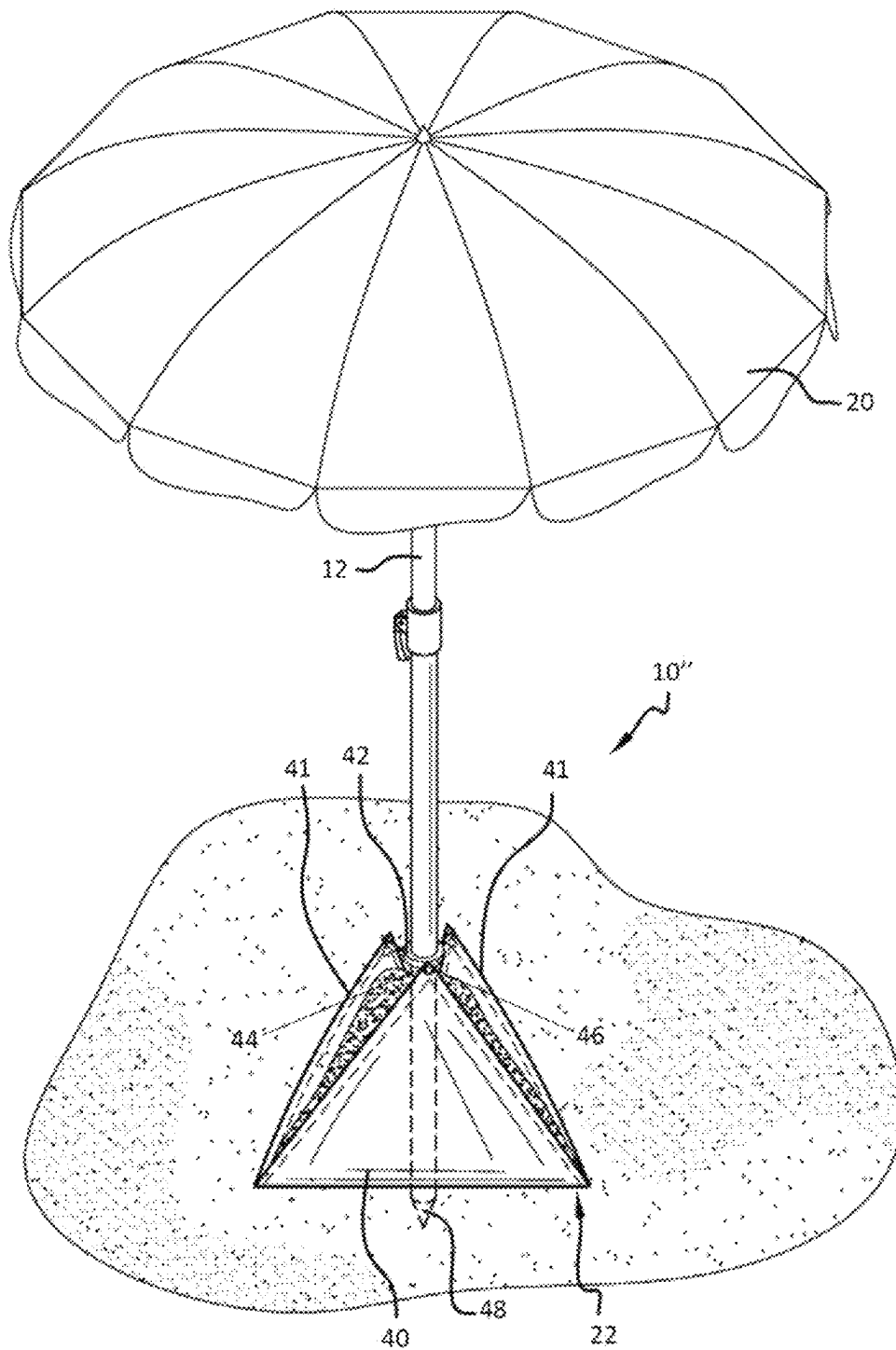


FIG. 3

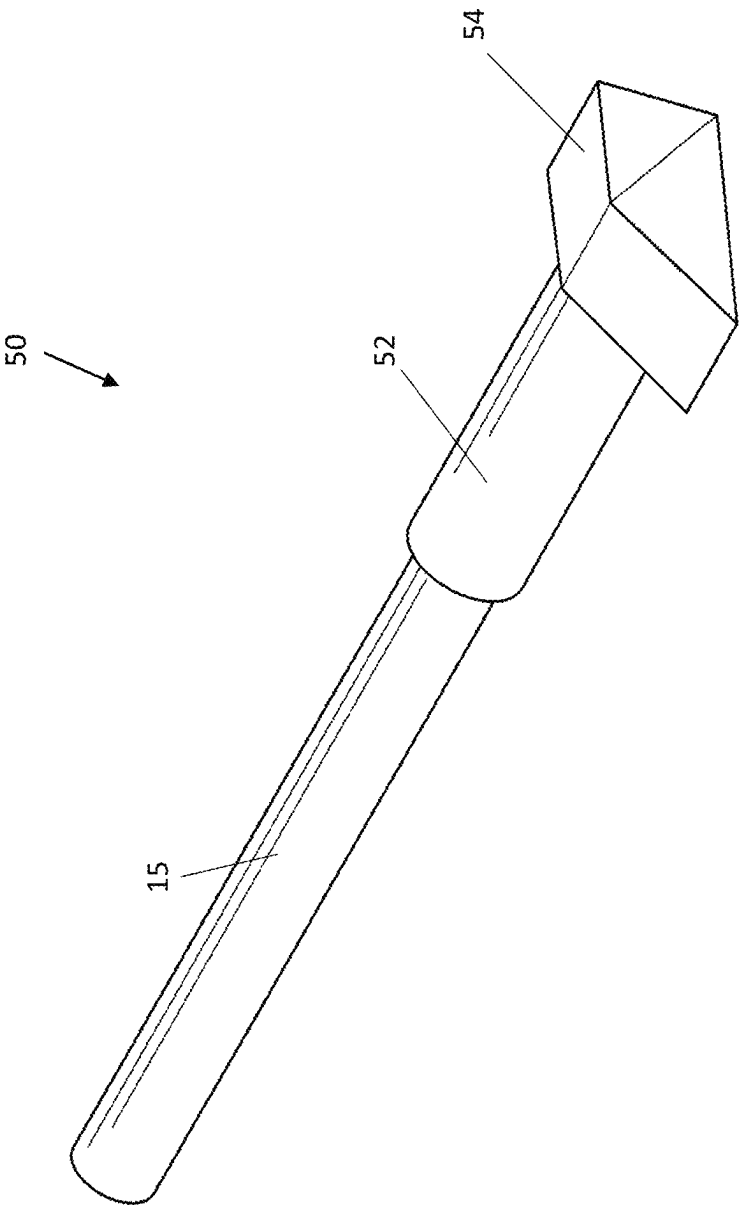


FIG. 4

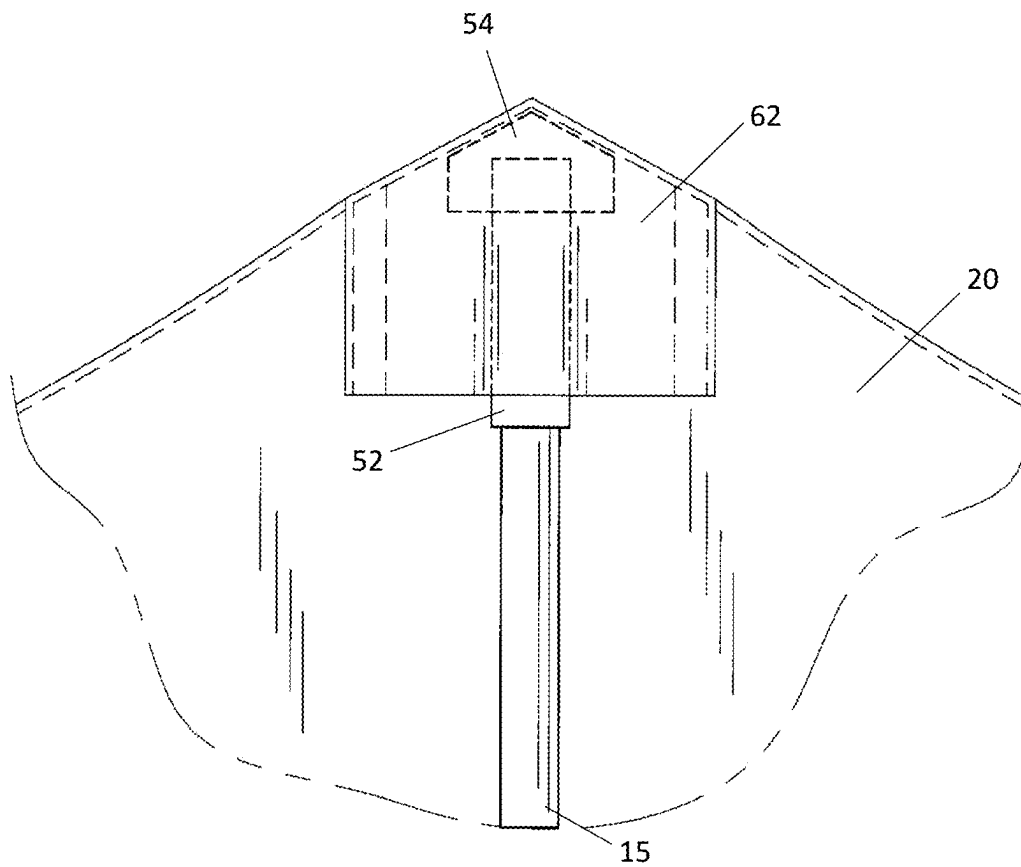


FIG. 5

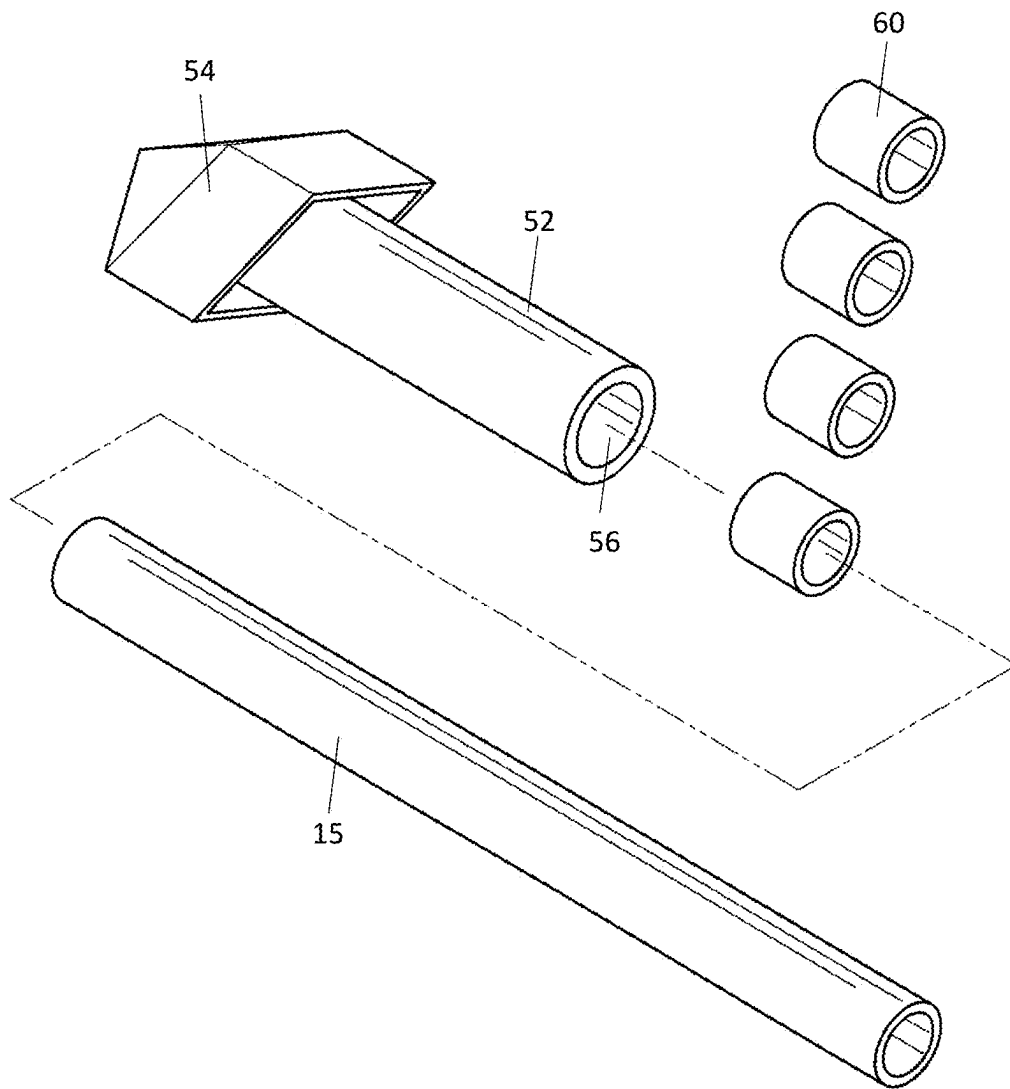


FIG. 6

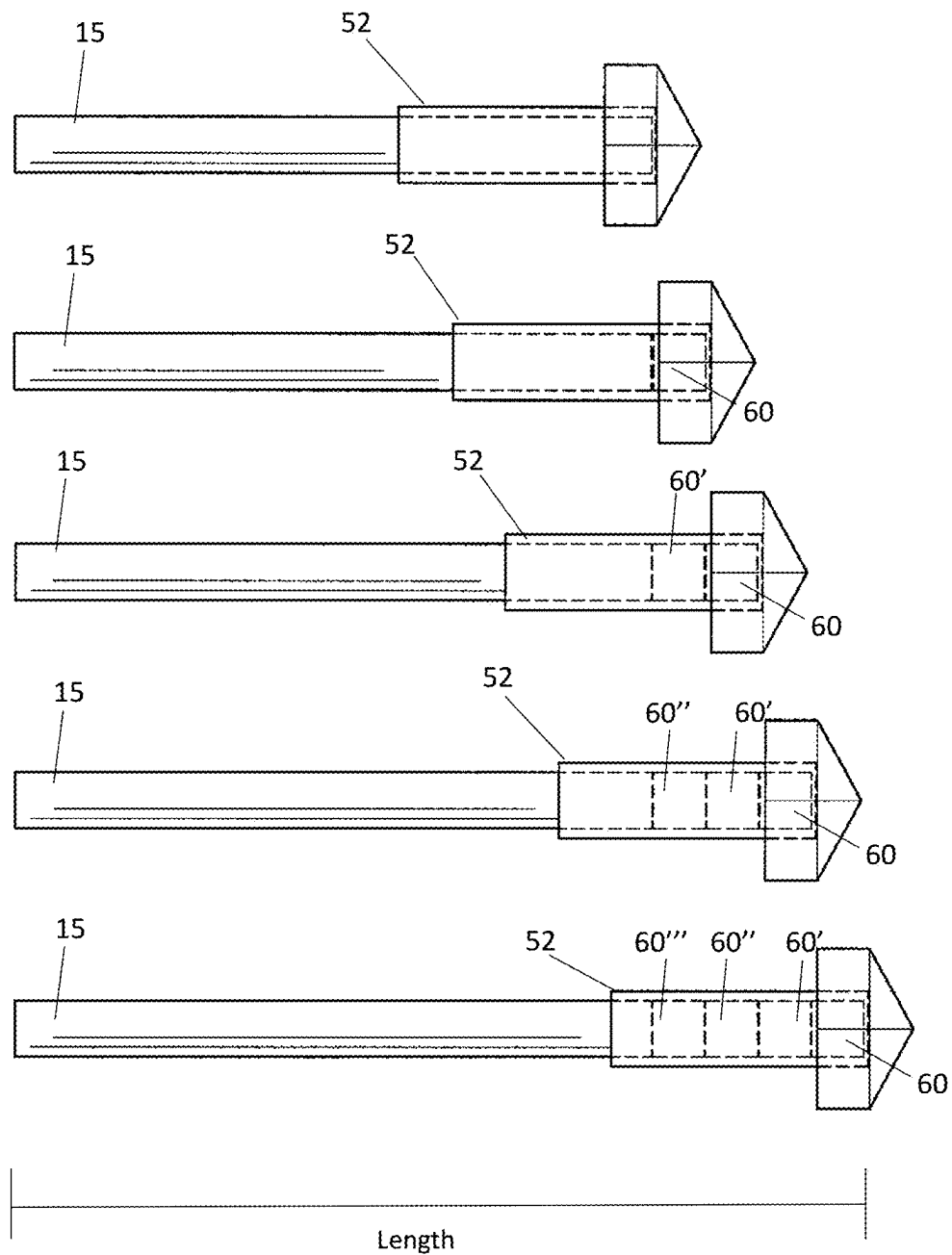


FIG. 7

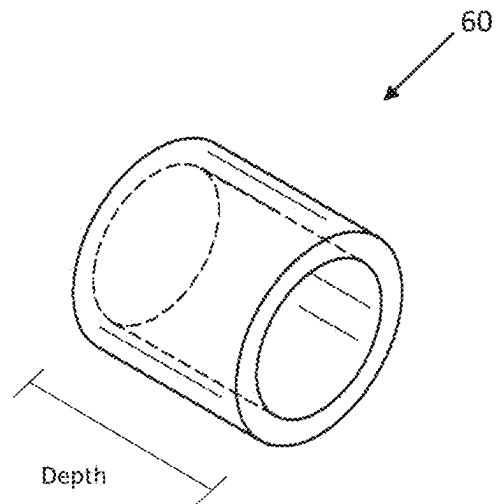


FIG. 8

1

VISOR DEVICE AND ASSEMBLY FOR AN UMBRELLA

This application claims the benefit of U.S. provisional application No. 63/189,360, filed May 17, 2021, which is incorporated herein by reference in its entirety.

FIELD OF THE TECHNOLOGY

The present inventions relate generally to umbrellas and, more particularly, to a visor system and umbrella assembly.

BACKGROUND

Umbrellas have long provided enjoyment and relaxation on sunny days for those seeking shade to avoid uncomfortable overheating and/or sunburn. Traditionally, sun shading umbrellas have a centrally disposed pole on which a collapsible sun shade is radially positioned.

Applicant desires reliable and user-friendly umbrella assemblies and systems that are useful and maintain the integrity of the shading shape in a variety of set-ups and environments.

SUMMARY

In accordance with the present disclosure, umbrella assemblies and systems are provided for achieving and maintaining improved shading capabilities and stabilization, for instance during wind events, and the like.

In one embodiment, an umbrella visor includes an upper frame; a lower frame; a pair of distal hinges hingedly connecting the upper frame about the lower frame; and an enclosure substantially spanning between the upper frame and the lower frame, and wherein the upper frame and the lower frame align together in a first position, and the lower frame being spaced from the upper frame in a second shielding position.

In certain examples, the upper frame includes an upper front member and at least one upper first offset member. The device may include an upper first proximate hinge between the upper front member and the at least one upper first offset member. The upper frame may include an upper second offset member. The upper front member, upper first offset member, and upper second offset member may hinge and align about one another in a storage position. The lower frame may include a lower front member and at least one lower first offset member. The device may include a lower first proximate hinge between the lower front member and the at least one lower first offset member. The lower frame may include a lower second offset member. The lower front member, lower first offset member, and lower second offset member may hinge and align about one another in a storage position. The upper frame and the lower frame may hinge and align about one another in a storage position. The upper frame may be parallel with the lower frame. The umbrella visor may be adapted to be removably attached to an umbrella. The umbrella visor may be telescopic to permit a height of the visor to be altered for effective shading.

In certain examples, the device may include at least one aperture. The device may include a fastener protruding through the aperture and connecting to an umbrella. The device may include a plurality of apertures and a plurality of fasteners. The device may include at least one aperture aligned about a front panel. The device may include at least one aperture aligned about a side panel. The enclosure may include a fabric. The enclosure may include a material

2

having a flexibility adapted to collapse upon itself, or the like. The enclosure may include a substantial rigidity. The enclosure may include a high ultraviolet protection factor. The device may include a front panel. The front panel may include a substantially rectangular shape. The device may include at least one side panel. The side panel may include a substantially triangular shape. The device may include two side panels. The device may include a front panel and a pair of opposing side panels. The device may include a pair of bends aligned between the front panel and respective adjacent side panels in an assembled position.

In one embodiment, an umbrella visor includes an upper front member and an upper offset member pivotally aligned about an upper proximate hinge; a lower front member and a lower offset member pivotally aligned about a lower proximate hinge; and a distal hinge pivotally connecting the upper offset member and the lower offset member.

In certain examples, the device may include an enclosure. The enclosure may include a fabric. The enclosure may include a material having a flexibility adapted to collapse upon itself. The enclosure may include a substantial rigidity. The enclosure may include a high ultraviolet protection factor.

In certain examples, the upper front member and upper offset member may hinge and align about one another in a storage position. The lower front member and lower offset member may hinge and align about one another in a storage position. The upper front member, upper offset member, lower front member, and lower offset member may hinge and align about one another in a storage position. The umbrella visor may be adapted to be removably attached to an umbrella. The umbrella visor may be telescopic to permit a height of the visor to be altered for effective shading. The device may include at least one aperture. The device may include a fastener protruding through the aperture and connecting to an umbrella. The device may include a plurality of apertures and a plurality of fasteners. The device may include at least one aperture aligned about a front panel. The device may include at least one aperture aligned about a side panel. The device may include a front panel. The front panel may include a substantially rectangular shape. The device may include at least one side panel. The side panel comprises a substantially triangular shape. The device may include two side panels. The device may include a front panel and a pair of opposing side panels. The device may include a pair of bends aligned between the front panel and respective adjacent side panels in an assembled position.

In one embodiment, an umbrella assembly comprises an umbrella having a canopy; and a visor assembly adapted to be removably attached to the canopy, and wherein the visor assembly being telescopic to permit a height of the visor assembly to be altered for effective shading.

In certain examples, the assembly may include an internal rib assembly. The internal rib assembly may include an upper frame and a lower frame. The upper frame may be hingedly connected about the lower frame. The internal rib assembly may include an upper front member and at least one upper first offset member. The assembly may include an upper first proximate hinge between the upper front member and the at least one upper first offset member. The internal rib assembly may include a lower front member and at least one lower first offset member. The assembly may include a lower first proximate hinge between the lower front member and the at least one lower first offset member. The internal rib assembly may include an upper front member and at least one upper second offset member. The assembly may include

3

an upper first proximate hinge between the upper front member and the at least one upper second offset member.

In certain examples, the internal rib assembly may include a lower front member and at least one lower second offset member. The assembly may include a lower second proximate hinge between the lower front member and the at least one lower second offset member. The internal rib assembly may include an upper first offset member, a lower first offset member, and a first distal hinge pivotally securing the upper first offset member about the lower first offset member. The internal rib assembly may include an upper second offset member, a lower second offset member, and a second distal hinge pivotally securing the upper second offset member about the lower second offset member.

In certain examples, the visor assembly may include a plurality of member tunnels adapted to secure the internal rib assembly. The visor assembly may include at least one aperture. The assembly may include a fastener protruding through the aperture and connecting to the umbrella. The assembly may include a plurality of apertures and a plurality of fasteners. The assembly may include at least one aperture aligned about a front panel. The assembly may include at least one aperture aligned about a side panel. The visor assembly may include an enclosure. The enclosure comprising a fabric. The enclosure may include a material having a flexibility adapted to collapse upon itself. The enclosure may include a substantial rigidity. The enclosure may include a substantial rigidity.

In certain examples, the visor assembly may be adapted to be positioned between at least two different locations along a perimeter of the canopy. The visor assembly may include a front panel. The front panel may include a substantially rectangular shape. The visor assembly may include at least one side panel. The side panel may include a substantially triangular shape. The visor assembly may include two side panels. The visor assembly may include a front panel and a pair of opposing side panels. The assembly may include a pair of bends aligned between the front panel and respective adjacent side panels. The assembly may include at least one securement wrap. The assembly may include a storage unit. The assembly may include a second visor assembly.

In certain examples, the assembly may include an end cap assembly to modify a tension of the canopy to prevent excessive or insufficient tension when the canopy is mounted by adjusting a depth of one or more channels before a distal end of a rib is inserted. The assembly may include a base installed on the bottom of a pole adapted for retaining the umbrella in an upright position. The base may include a tarp including a reinforcement disposed centrally within the tarp, an opening disposed within the reinforcement, a plurality of spaced apart grommets disposed outwardly within the tarp, whereby each grommet is adapted to be attached to the pole.

In one embodiment, an umbrella assembly comprises an umbrella having a canopy; a visor assembly having a front panel and at least one side panel adjacent the front panel, and adapted to align along a perimeter of the canopy; a front air gap vent between the front panel and a portion of the canopy perimeter; and a side air gap vent between the at least one side panel and a portion of the canopy perimeter, and wherein the front air gap vent and the side air gap vent adapted to stabilize the umbrella in a wind event.

In certain examples, the at least one side panel may include a distal end angled from a proximate end. The visor assembly may removably attach and/or detach to the canopy. The visor assembly may be telescopic to permit a height of the visor assembly to be altered for effective shading. The

4

assembly may include an internal rib assembly. The internal rib assembly may include an upper frame and a lower frame. The upper frame may be hingedly connected about the lower frame. The internal rib assembly may include an upper front member and at least one upper first offset member. The assembly may include an upper first proximate hinge between the upper front member and the at least one upper first offset member. The internal rib assembly may include a lower front member and at least one lower first offset member. The visor assembly may include a plurality of member tunnels adapted to secure the internal rib assembly.

In certain examples, the visor assembly may include at least one aperture. The assembly may include a fastener protruding through the aperture and connecting to the umbrella. The assembly, a plurality of apertures and a plurality of fasteners. The assembly may include at least one aperture aligned along the front panel. The assembly may include at least one aperture aligned along the side panel. The front panel may include a substantially rectangular shape. The side panel may include a substantially triangular shape. The assembly may include at least one securement wrap. The assembly may include an end cap assembly adapted to modify a tension of the canopy to prevent excessive or insufficient tension when the canopy is mounted by adjusting a depth of one or more channels before a distal end of a rib is inserted. The assembly may include a base installed on the bottom of a pole adapted for retaining the umbrella in an upright position. The base may be a tarp including a reinforcement disposed centrally within the tarp, an opening disposed within the reinforcement, a plurality of spaced apart grommets disposed outwardly within the tarp, whereby each grommet is adapted to be attached to the pole.

In one embodiment, an umbrella assembly includes a pole having a rib assembly at one end comprising a plurality of ribs aligning a canopy, and a distal finial plug; and a base.

In certain examples, the assembly includes a flagstaff. The finial plug may removably mount a lower length of the flagstaff. The flagstaff may include a proximate flag connection. The proximate flag connection may include at least one proximate connector. The flagstaff may include a distal flag connection. The distal flag connection may include at least one distal connector. The finial plug may have an outer diameter sidewall extending from a retaining base and an aperture. The finial plug may include an outer diameter sidewall extending from a retaining base, an inner diameter sidewall extending from the outer diameter sidewall, and an aperture recessed within the inner diameter sidewall.

In particular examples, the canopy includes a channel having an upper region and a lower region, wherein the finial plug adapted to align along the upper region. The channel may include at least one alignment member. The channel may include a pair of opposing alignment members. The finial plug may align within the channel above the alignment member. The lower length of the flagstaff may align below a mounting surface in an assembled position. The flagstaff may extend from the finial plug independent of an opened extended canopy position and a closed retracted position.

In certain examples, the assembly includes rib assembly comprising a plurality of ribs. The rib assembly may include an end cap assembly attached to a distal end of each rib, wherein the end cap assembly including a channel with an adjustable depth for receiving the rib. The end cap assemblies may modify the tension of the canopy to prevent excessive or insufficient tension when the canopy is mounted by adjusting a depth of one or more channels before the distal end of the rib is inserted.

5

In some examples, the base may be installed on the bottom of the pole for retaining the umbrella in an upright position. The base may include a tarp including a reinforcement disposed centrally within the tarp, an opening disposed within the reinforcement, a plurality of spaced apart grommets disposed outwardly within the tarp, whereby each grommet is adapted to be attached to the pole.

In particular examples, the assembly includes a hub for the rib assembly to attach to the pole. The assembly may include pulley assembly to open and close the canopy and a pin adapted to be inserted into a hole in the pole to retain the canopy in an open configuration. The assembly may include a crank assembly adapted to open and close the canopy. The assembly may include a tilt mechanism attached onto the pole to tilt the rib assembly and canopy at an angle with respect to the pole. The assembly may include a vent on the canopy for air to pass through the canopy. The assembly may include one or more inserts adapted to fit within the channel of the tube to modify the tension of the canopy by extending an effective length of the rib of the umbrella frame. For instance, the tension of the canopy can be increased by adding inserts into the tube to increase the effective length of the rib that the end cap assembly is attached to and the tension of the canopy can be decreased by removing inserts from the tube to decrease the effective length of the rib that the end cap assembly is attached thereto.

In another embodiment, an umbrella assembly includes an umbrella with a radially extending canopy and a finial plug; and a flagstaff with multi flag alignment connector fasteners.

In yet another embodiment, an umbrella assembly includes an umbrella having a finial plug secured along a distal portion of a canopy; a flagstaff having a diameter sized to mate within the finial plug, and wherein the flagstaff being removably mounted in a substantially vertical position within the finial plug.

In some examples, the umbrella comprising a central pole having a head portion. The head portion may include a channel having an upper region and a lower region, wherein the finial plug adapted to align along the upper region. The channel may include at least one alignment member. The channel may include a pair of opposing alignment members. The finial plug may align within the channel above the alignment member. The finial plug may removably mount a lower length of the flagstaff. The finial plug may have an outer diameter sidewall extending from a retaining base and an aperture. The finial plug may include an outer diameter sidewall extending from a retaining base, an inner diameter sidewall extending from the outer diameter sidewall, and an aperture recessed within the inner diameter sidewall. The outer diameter sidewall may be sized to complement a channel for receiving and retaining the finial plug.

In certain examples, the lower length of the flagstaff is aligned below a mounting surface in an assembled position. The flagstaff may extend from the finial plug independent of an opened extended canopy position and a closed retracted position. The umbrella may include a rib assembly comprising a plurality of ribs. The umbrella may include a rib assembly comprising an end cap assembly attached to a distal end of each rib, wherein the end cap assembly including a channel with an adjustable depth for receiving the rib. The end cap assemblies may modify the tension of the canopy to prevent excessive or insufficient tension when the canopy is mounted by adjusting a depth of one or more channels before the distal end of the rib is inserted.

In particular examples, the assembly includes a base installed on the bottom of the pole adapted for retaining the umbrella in an upright position. The base may include a tarp

6

including a reinforcement disposed centrally within the tarp, an opening disposed within the reinforcement, a plurality of spaced apart grommets disposed outwardly within the tarp, whereby each grommet is adapted to be attached to the pole.

In some examples, the assembly may include a hub for the rib assembly to attach to the pole. The assembly may include a pulley assembly to open and close the canopy and a pin adapted to be inserted into a hole in the pole to retain the canopy in an open configuration. The assembly may include a crank assembly adapted to open and close the canopy. The assembly may include a tilt mechanism attached onto the pole to tilt the rib assembly and canopy at an angle with respect to the pole. The assembly may include a vent on the canopy for air to pass through the canopy. The assembly may include one or more inserts adapted to fit within the channel of the tube to modify the tension of the canopy by extending an effective length of the rib of the umbrella frame. For instance, the tension of the canopy can be increased by adding inserts into the tube to increase the effective length of the rib that the end cap assembly is attached to and the tension of the canopy can be decreased by removing inserts from the tube to decrease the effective length of the rib that the end cap assembly is attached thereto.

In another embodiment, an umbrella system includes a central pole having a head portion; a plurality of radially depending ribs pivotally mounted adjacent the head portion supporting a canopy; and an aperture on a distal end of the central pole above the head portion.

In some examples, the aperture comprises a finial plug. The finial plug may align within a channel above an alignment member. The finial plug may removably mount a lower length of a flagstaff. The finial plug may include an outer diameter sidewall extending from a retaining base and an aperture. The finial plug may include an outer diameter sidewall extending from a retaining base, an inner diameter sidewall extending from the outer diameter sidewall, and an aperture recessed within the inner diameter sidewall.

The system may include a flagstaff. The flagstaff may include a proximate flag connection. The proximate flag connection may include at least one proximate fastener. The flagstaff may include a distal flag connection. The distal flag connection may include at least one distal fastener.

In one embodiment, an umbrella comprises a frame having a pole with a rib assembly at one end, the rib assembly comprising a plurality of ribs; an end cap assembly attached to a distal end of each rib, the end cap assembly including a channel with an adjustable depth for receiving the rib; and a canopy attached to the end cap assembly of each rib of the frame. The end cap assemblies are adapted to modify the tension of the canopy to prevent excessive or insufficient tension when the canopy is mounted by adjusting the depth of one or more channels before the distal end of the rib is inserted. The end cap assembly may be tubular.

The umbrella may further include a base installed on the bottom of the pole adapted for retaining the umbrella in an upright position. For example, the base may be comprised of a rigid container adapted for receiving a fluid. In another example, the base may be a tarp including a reinforcement disposed centrally within the tarp, an opening disposed within the reinforcement, a plurality of spaced apart grommets disposed outwardly within the tarp, whereby each grommet is adapted to be attached to the pole.

One embodiment of the umbrella includes a hub for the rib assembly to attach to the pole. A pulley assembly may be included to open and close the canopy and further include a pin to be inserted into a hole in the pole to retain the canopy in an open configuration. In another example, a crank

assembly may be included to open and close the canopy. The umbrella may also include a tilt mechanism attached onto the pole to tilt the rib assembly and canopy at an angle with respect to the pole.

A finial may be added to the top of the canopy. The canopy may further include a vent for air to pass through the canopy.

Another aspect of the present invention is directed to an end cap assembly adapted to be installed onto a distal end of a rib for an umbrella frame to modify the tension of a canopy attached to the umbrella frame. The end cap assembly comprises a tube having an open end with a channel adapted to receive a rib of an umbrella frame and a closed end adapted to be received by a canopy; and one or more inserts adapted to fit within the channel of the tube to modify the tension of the canopy by extending an effective length of the rib of the umbrella frame. The tension of the canopy can be increased by adding inserts into the tube to increase the effective length of the rib that the end cap assembly is attached to and the tension of the canopy can be decreased by removing inserts from the tube to decrease the effective length of the rib that the end cap assembly is attached to.

In one embodiment, the closed end of the tube is substantially T-shaped.

The insert may be comprised of rubber. Between about one and about four inserts may be inserted into the tube.

In some examples, the length of the tube may range between about 1 inch to about 5 inches. For instance, the length of the tube may be about 2.5 inches. In some examples, the depth of the insert is between about 5 millimeters and about 20 millimeters. For instance, the depth of the insert may be about 10 millimeters.

Still another aspect of the present invention is directed to an umbrella comprising a frame having a pole with a rib assembly at one end, the rib assembly comprising a plurality of ribs; an end cap assembly attached to a distal end of each rib to modify the tension of a canopy attached to the umbrella frame; and a canopy attached to the end cap assembly of each rib of the frame. The end cap assembly comprises a tube having an open end with a channel adapted to receive a rib of an umbrella frame and a closed end adapted to be received by a canopy; and one or more inserts adapted to fit within the channel of the tube to modify the tension of the canopy by extending an effective length of the rib of the umbrella frame. The tension of the canopy can be increased by adding inserts into the tube to increase the effective length of the rib that the end cap assembly is attached to and the tension of the canopy can be decreased by removing inserts from the tube to decrease the effective length of the rib that the end cap assembly is attached to.

The present invention may also be considered a method for adjusting the canopy tension of an umbrella. In one example, each rib is fitted with a "T" shape tubular end cap. The cap may be loosely fitted so that it can be easily removed without the use of any tools. A small insert may then be added to each cap before it is attached to the rib. By doing so, the rib has been effectively lengthened by the size of the insert. After all end caps have been attached to the ribs, the canopy is attached and the umbrella opened and checked for tension. If the tension is too loose, then additional inserts may be added effectively lengthening each rib. Conversely, if the canopy is too tight then inserts can be removed effectively reducing the length of the rib.

The above summary was intended to summarize certain embodiments of the present disclosure. Embodiments will be set forth in more detail in the figures and description of embodiments below. It will be apparent, however, that the description of embodiments is not intended to limit the

present inventions, the scope of which should be properly determined by the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the disclosure will be better understood by a reading of the Description of Embodiments along with a review of the drawings, in which:

FIG. 1 is a front perspective view of an umbrella visor assembly according to one embodiment of the present invention;

FIG. 1a is a front perspective view of an umbrella assembly according to one embodiment of the present invention, with elements removed for clarity;

FIG. 1b is a front perspective view of an umbrella visor in an assembled position according to one embodiment of the present invention;

FIG. 1c is a perspective view of isolated elements shown in FIG. 1b;

FIG. 1d is a top view of isolated frame elements incorporated into one embodiment of the present invention introduced in FIG. 1b;

FIG. 1e is a top perspective view of isolated elements shown in FIG. 1d in a storage position;

FIG. 1f is a side perspective view of isolated elements shown in FIG. 1b;

FIG. 1g is a rear perspective view of an isolated elements introduced in FIG. 1b;

FIG. 1h is a side perspective view of isolated visor elements according to one embodiment of the present invention;

FIG. 2 is a front perspective view of an umbrella according to one embodiment;

FIG. 3 is a top perspective view of an umbrella according to another embodiment;

FIG. 4 is a top perspective view of an end cap assembly according to one embodiment;

FIG. 5 is an enlarged cross-sectional view of an end cap assembly installed over a rib inserted into a canopy;

FIG. 6 is a disassembled view of the end cap assembly shown in FIG. 4;

FIG. 7 is a partially cross-sectional view of the end cap assembly shown in FIG. 4; and

FIG. 8 is a side perspective view of an insert according to one embodiment of the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

In the following description, like reference characters designate like or corresponding parts throughout the several views. Also in the following description, it is to be understood that such terms as "forward," "rearward," "left," "right," "upwardly," "downwardly," and the like are words of convenience and are not to be construed as limiting terms.

Referring now to the drawings in general and FIG. 1 in particular, it will be understood that the illustrations are for the purpose of describing an embodiment of the invention and are not intended to limit the inventions thereto.

As illustrated in FIGS. 1b, 1c, 1f-1h, one embodiment of an umbrella assembly includes an umbrella having a canopy 20; and a visor assembly 80 that is removably attached to the canopy in a variety of positions as shown and described herein and is movable, including telescopic and the like, to permit a height of the visor assembly to be altered for

effective shading. Visor assembly **80** when not in use can be stored in bag **140**. Visor assembly **80** also includes tie-backs **108** as seen in FIG. 1.

In one embodiment, an umbrella visor includes an upper frame; a lower frame; a pair of distal hinges **124**, **126** hingedly connecting the upper frame about the lower frame; and an enclosure **106** substantially spanning between the upper frame and the lower frame. The upper frame and the lower frame generally align together in a first position, i.e. any of the non-blocking positions shown and described herein, and the lower frame is generally spaced from the upper frame in any of the second shielding position shown and described herein.

In certain examples, the upper frame includes an upper front member **112** and at least one upper first offset member **116**. The device may include an upper first proximate hinge **132** between the upper front member **112** and the at least one upper first offset member **116**. The upper frame may include an upper second offset member **120**. The upper front member **112**, upper first offset member **116**, and upper second offset **120** member may hinge and align about one another in a storage position, as shown and described herein, for instance in FIG. 1e. The lower frame may include a lower front member **114** and at least one lower first offset member **118**. The device may include a lower first proximate hinge **134** between the lower front member **114** and the at least one lower first offset member **118**. The lower frame may include a lower second offset member **122**. The lower front member **114**, lower first offset member **118**, and lower second offset member **122** may hinge and align about one another in a storage position, as shown and described herein, for instance in FIG. 1e. Similarly, the upper frame and the lower frame may hinge and align about one another in a storage position. For instance, the upper frame may be parallel, or similar thereto, with the lower frame.

The umbrella visor **80** may be removably attached to an umbrella in a variety of ways and may be telescopic to permit a height of the visor to be altered for effective shading as shown and described herein. In certain examples, the device **80** may include at least one aperture **94** (as shown in FIG. 1c). The device may include a fastener **88** protruding through the aperture **94** and connecting to umbrella in a variety of ways and combinations. The device may include a plurality of apertures **94** and a plurality of fasteners **88**. For instance, at least one aperture **94** aligned about a front panel **86**, at least one aperture **94** aligned about a side panel **84**, and any combination thereof.

Various shielding materials, shapes, and constructions may be incorporated into the visor **80**. For instance, the enclosure **106** may include a fabric, any material having a flexibility to collapse upon itself, or the like, a variety of rigidity, a variety of ultraviolet protection factors, and the like.

As shown in the various figures, the front panel **86** may include a substantially rectangular shape, while those skilled in the art having the benefit of this disclosure will recognize additional and other shapes. Further, the side panel **84** may include a substantially triangular shape, while those skilled in the art having the benefit of this disclosure will recognize additional and other shapes. As illustrated, the assembled device may include a pair of bends **82** aligned between the front panel and respective adjacent side panel(s), for instance to match the perimeter, or similar, of a canopy **20**.

The assembly may include a variety of internal rib frame assemblies **110**. For instance, in one embodiment, an umbrella visor **80** includes an upper front member **112** and an upper offset member **116**, **120** pivotally aligned about

upper proximate hinges **128**, **132**; a lower front member **114** and lower offset members **118**, **122** pivotally aligned about lower proximate hinges **130**, **134**; and a distal hinges **124**, **126** pivotally connecting the upper offset members **116**, **120** and the lower offset members **118**, **122**.

In certain examples, the visor assembly may include a plurality of member tunnels **92** to secure any of the internal rib assemblies shown and described herein.

In one embodiment, an umbrella assembly comprises an umbrella having a canopy **20**; a visor assembly **80** having a front panel **86** and at least one side panel **84** that is generally adjacent the front panel **86** to align along a perimeter of the canopy **20**. As illustrated in FIG. 1b, a front air gap vent **102** is between the front panel **86** and a portion of the canopy perimeter **20**; and a side air gap vent **104** is between the at least one side panel **84** and a portion of the canopy perimeter **20**. Applicant has unexpectedly discovered unforeseen stabilization advantages created by the visor **80**. For instance, during wind events, and the like, the visor **80** creates a stabilized air foil for enhanced stability performance.

FIG. 1a introduces flagstaff assembly **70** that is removably mounted about umbrella assembly **100** and is comprised of a pole **12** having a rib assembly **14** to support a canopy **20**. One or more ribs **15** includes an end cap assembly at its distal end, wherein each end cap assembly is inserted into the canopy **20** to mount it onto the rib assembly **14**. As shown, the flagstaff assembly **70** generally includes at least one flag **76**, but including two or more as shown and described herein, secured to flagstaff **74** at connector fasteners. Flagstaff **74** is generally removably mounted in a substantially vertical position within an aperture, finial plug **30**, or the like in the upper portion of the pole **12** or canopy **20**.

In certain examples, the finial plug **30** is secured along a distal portion of canopy **20**, and includes aperture **72** to receive and retain any of the flagstaffs shown and described herein. Further, the canopy **20** may have a channel in opening adapted to receive and retain a corresponding finial plug or the like. The opening may be aligned along a substantially flat mating surface. The channel may have an upper region and a lower region, thus the finial plug may align along the upper region. As shown, channel may have one, including at least a pair of opposing, alignment members. In certain examples, finial plug **30** may be sized for snug slide mounting downwardly into channel, for instance to telescopically slide into channel. In particular examples, the outer diameter sidewall is sized to snugly telescopically slide into channel about alignment members to ease movement of the pole within the distal portion of the canopy as shown and described herein.

Those of ordinary skill in the art having the benefit of this disclosure will recognize numerous connection and arrangement configurations between the umbrella canopy/top end and the corresponding flagstaff. As shown in certain embodiments, finial plug **30** has an outer diameter sidewall extending from a retaining base to support aperture **72**. In certain examples, finial plug **30** has an outer diameter sidewall extending from retaining base, an inner diameter sidewall **92** extending from outer diameter sidewall (or the like), with aperture **72** recessed within inner diameter sidewall **92**.

The embodiments and examples herein support one, two, and a plurality of flags, and the like, above the umbrella in a variety of configurations. The flagstaff **74** includes connectors to mount flag **76** in a semi-fixed position to flagstaff **74**. As shown flagstaff **74** includes a distal flag connection **78** to secure a first flag (or the like) and a proximate flag connection **78'** to secure at least a second flag (or the like).

11

The distal flag connection **78** includes at least one distal connection fastener to secure flag **76**, or the like. Further, in certain examples, the proximate flag connection **78'** includes at least one proximate connection fastener to secure a second or additional flag, or the like. In certain examples, flagstaff **74** may be generally cylindrical and elongated along an axis. Flag **76** may be repositionable about flagstaff **74** as shown and described herein.

As shown in the various figures, pole **12** may be supported by a base **22**. For instance, the base **22** may comprise a rigid container with sand, water or another fluid inside. The rib assembly **14** may be attached to the pole **12** via a hub **24**. When the hub **24** is raised along the pole **12**, the canopy **20** is opened. As the hub **24** is lowered along the pole **12**, the canopy **20** is closed. A pulley assembly **26** may be included to raise and lower the hub **24**. The hub **24** may be held in its position using a pin **27**.

The canopy **20** may also include a vent **32** for air to pass through the canopy in order to enable heat to escape from underneath the umbrella and allow cool air to enter under the umbrella.

Another example of an umbrella **10'** having a canopy tension system is shown in FIG. 2. The umbrella includes a crank assembly **28** adapted to open and close the canopy, and further includes a tilt mechanism **34** for angling the canopy in a desired direction.

FIG. 3 illustrates another example of an umbrella **10"** with a canopy tension system. The umbrella **10"** is supported by a base **22**. In this example, the base **22** is comprised of a tarp **40** adapted to attach to a collar **42** installed onto the pole **12**. In one embodiment, the tarp **40** is comprised of three outer triangles that attach to the collar **42**. Each outer triangle **41** may be attached to the collar **42** via a grommet **44** that engages with a fastener **46** on the collar **42**. Other similar mechanisms may also be used to attach the tarp **40** to the pole **12**, including fasteners installed directly onto each outer triangle **41** that are inserted into holes on the pole **12** or collar **42**. The tarp **40** also includes a reinforcement with an opening for the end **48** of pole **12** to insert through and into the ground. Sand may be used to fill the base **22**.

One example of an end cap assembly **50** to be used as a canopy tension system is shown in FIGS. 4-6. The end cap assembly **50** is comprised of a tube **52** adapted to receive a distal end of rib **15** (wherein the proximal end of rib **15** is attached to the hub **24**). The tube **52** may further include a T-shaped end **54** adapted to disperse pressure on the canopy when inserted into pocket **62** of the canopy **20** to prevent the tube **52** from slipping out.

FIG. 6 provides a partially disassembled view of the end cap assembly **50** wherein a plurality of inserts **60** may be inserted into a channel **56** prior to insertion of the rib **15** into the channel **56**. FIG. 7 illustrates the effect of adding one or more inserts **60**, **60'**, **60"**, **60'''** into the tube **52**. As each insert **60** is added to the tube **52**, the depth of the channel **56** decreases for the rib **15** to enter into. By decreasing the depth for the rib **15** to enter, the effective length of the rib **15** increases when inserted into the pocket **62** of the canopy **20**. In certain examples, the length of the tube **52** is within a range wherein inserting the rib **15** into the end cap assembly **50** still results in an overall length wherein the canopy **20** can be mounted onto the rib assembly **14**. For example, the tube may have a length within a range of about 1 inch to about 5 inches. In one embodiment, the tube **52** may have a length of about 2.5 inches.

One example of an insert **60** adapted to be inserted into the tube **52** is shown in FIG. 8. The depth of each insert **60** is preferably within a range that is large enough to provide

12

a discernible effect on the tension of a canopy **20** with each insert added/removed, while at the same time, is small enough so that the tension of the canopy **20** can be adequately fine-tuned (i.e., adding or removing an insert **60** does not cause such an increase/decrease in canopy tension that the tension cannot be properly calibrated). For example, the depth of the insert may range between about 5 millimeters and about 20 millimeters. In one embodiment, the depth of the insert is about 10 millimeters, wherein each insert **60** provides about an additional 0.25 lb of tension to the canopy.

In operation, the end cap assembly may be used to adjust the tension of a canopy by modifying the effective length of the ribs to be inserted into the canopy. The initial tension may be determined by the size of the frame and the size of the canopy. During manufacturing of the umbrella itself, there are 9 possible outcomes that determine the tension of the canopy. The frame and canopy can be too small, just right or too large. When the canopy is attached to the frame, the statistical probability that the tension will be within an acceptable range is as follows:

- 1) Loose Canopy: $F1+C2$ or $F2+C3$
 - 2) Tight Canopy: $F2+C1$ or $F3+C2$
 - 3) Extremely Loose: $F1+C3$
 - 4) Extremely Tight: $F3+C1$
 - 5) Within acceptable range: $F1+C1$ or $F2+C2$ or $F3+C3$
- wherein $F1$ =Frame too small; $F2$ =Frame acceptable; $F3$ =Frame too large; $C1$ =Canopy too small; $C2$ =Canopy acceptable; $C3$ =Canopy too large.

The end cap assembly provides a solution for examples 1-4, wherein the frame and canopy have dimensions that are not within an initially acceptable range. For umbrellas that have a frame and canopy combination resulting in a loose canopy, the end cap assembly can be added to one or more ribs along with one or more inserts to increase the size of the frame to increase the tension within an acceptable range. For umbrellas that have a frame and canopy combination resulting in a tight canopy, the ribs may be initially shortened to reduce canopy tension, and an end cap assembly may be added to fine-tune the effective length of the rib to create the desired canopy tension. Moreover, as the canopy loosens over time, additional inserts may be added to the end cap assembly to maintain proper canopy tension. If the canopy tension becomes too tight, one or more inserts may be removed from the tube.

Certain modifications and improvements will occur to those skilled in the art upon a reading of the foregoing description. It should be understood that all such modifications and improvements have been deleted herein for the sake of conciseness and readability but are properly within the scope of the following claims.

What is claimed:

1. An umbrella assembly comprising:

- an umbrella including a canopy supported by a plurality of ribs;
- a visor assembly adapted to be removably attached to said canopy with a fastener protruding through an aperture defined through the visor assembly, said visor assembly including a front panel and a pair of opposing side panels; and
- an internal rib assembly positioned within said visor assembly and including an upper frame and a lower frame, wherein said upper frame is hingedly connected about said lower frame, wherein said upper frame includes an upper front member and at least one upper offset member and said lower frame includes a lower front member and at least one lower offset member, wherein said internal rib assembly further includes an

13

upper first offset member, a lower first offset member, and a first distal hinge pivotally securing said upper first offset member about said lower first offset member, wherein said internal rib assembly includes an upper second offset member, a lower second offset member, and a second distal hinge pivotally securing said upper second offset member about said lower second offset member; and

wherein said visor assembly is configured to be deployed in at least two configurations: a first configuration in which the visor assembly depends from a proximate end of at least one rib of the plurality of ribs supporting the canopy, and a second configuration wherein the visor assembly defines a smaller overall size to facilitate storage, the visor assembly additionally being telescopic to permit a height of said visor assembly to be altered for effective shading.

2. The assembly of claim 1, wherein said visor assembly has a plurality of member tunnels adapted to secure said internal rib assembly.

3. The assembly of claim 1, wherein said visor assembly has an enclosure.

14

4. An umbrella visor comprising:

an upper front member and an upper offset member pivotally aligned about an upper proximate hinge;

a lower front member and a lower offset member pivotally aligned about a lower proximate hinge; and

a distal hinge pivotally connecting said upper offset member and said lower offset member, and

wherein said umbrella visor is configured to be deployed in at least two configurations: a first configuration in which a visor assembly depends from a proximate end of a rib supporting the canopy via a fastener protruding through an aperture defined in the visor assembly, and a second configuration wherein the visor assembly defines a smaller overall size to facilitate storage, the visor assembly additionally being telescopic to permit a height of said visor to be altered for effective shading, and

wherein said umbrella visor is adapted to hinge and align about itself in a storage position.

* * * * *